COM 5506
Social Network Analysis for Communication

L6 Strength of weak ties

SEM B 2026

Fei Shen, City University of Hong Kong

Housekeeping

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- Schedule update

Each group consists 5-7 students.

7	3/4	Lecture: Strength of weak ties Lab: Textual analysis example	Deadline for grouping – send to TA
8	3/11	Lecture: Community detection Lab: Gephi introduction Workshop 1: project proposal consultation	EK Chap 16, TK Chap 5
9	3/18	Lecture: Network data hypothesis testing Lab: UCINET introduction	EK Chap 19, TK Chap 6 Bring your topics and ideas to class
10	3/25	Workshop 2: Group project progress consultation	Bring your data set to class
11	4/1	**Reading week**	
12	4/8	Final test (Lecture 1 to Lecture 8) Workshop 3: Group project finding consultation	Bring your preliminary findings to class
13	4/15	**No Class**	
14	4/22	*Final Project Presentation Day*	Complete your project draft
	4/29		**Research paper due**

Business trip cancelled. Three options for your choice:

1. move final test here.
2. move workshop here.
3. move both here and end class early on April 8 as presentation day.

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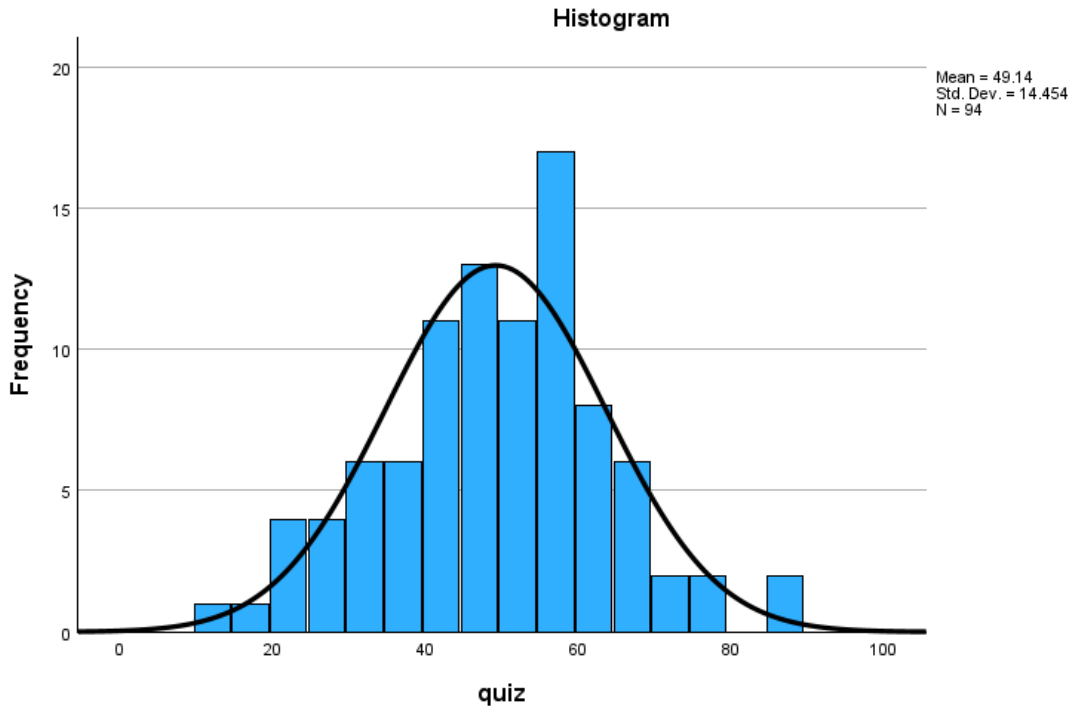
- Proposal format (due next week)
  - Within 1 page (preferably 2-3 ideas, with priority)
  - Group members
  - Your research topic
  - Your data collection method
  - Roughly how large is your data
  - Your main questions
  - I will discuss with each group next week
- This is all for your own reference and sorting our your thoughts. You don’t have to email me this. But I will discuss with your group during class.

Quiz

- Range 12-85 (last year 20-85)
- Mean: 49 (last year 52)
- SD: 14 (last year 9)
- Final exam, similar format



Our Journey Path

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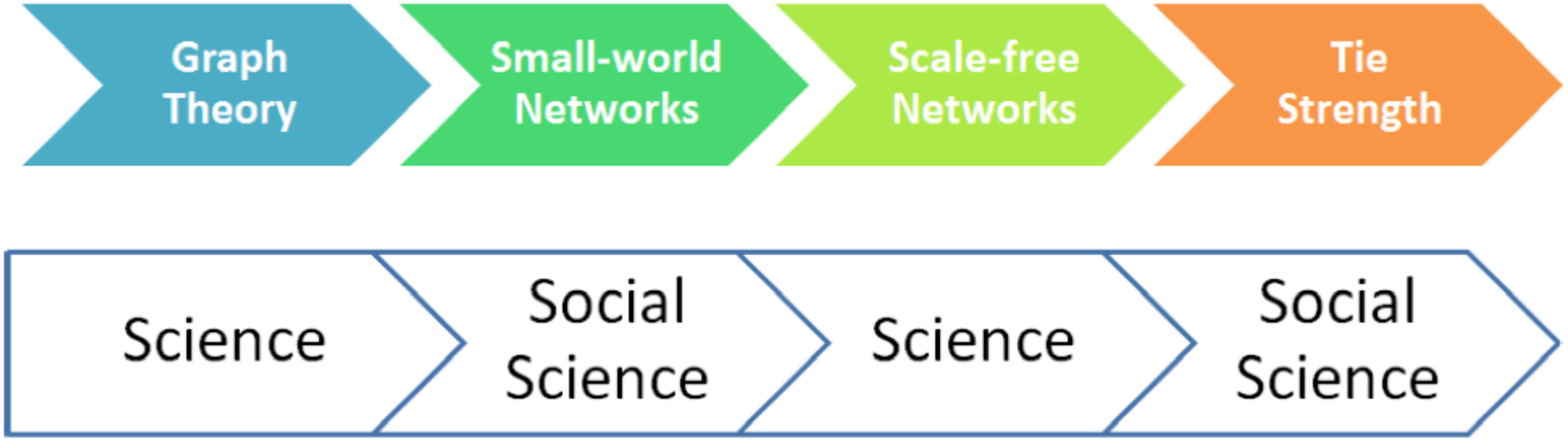
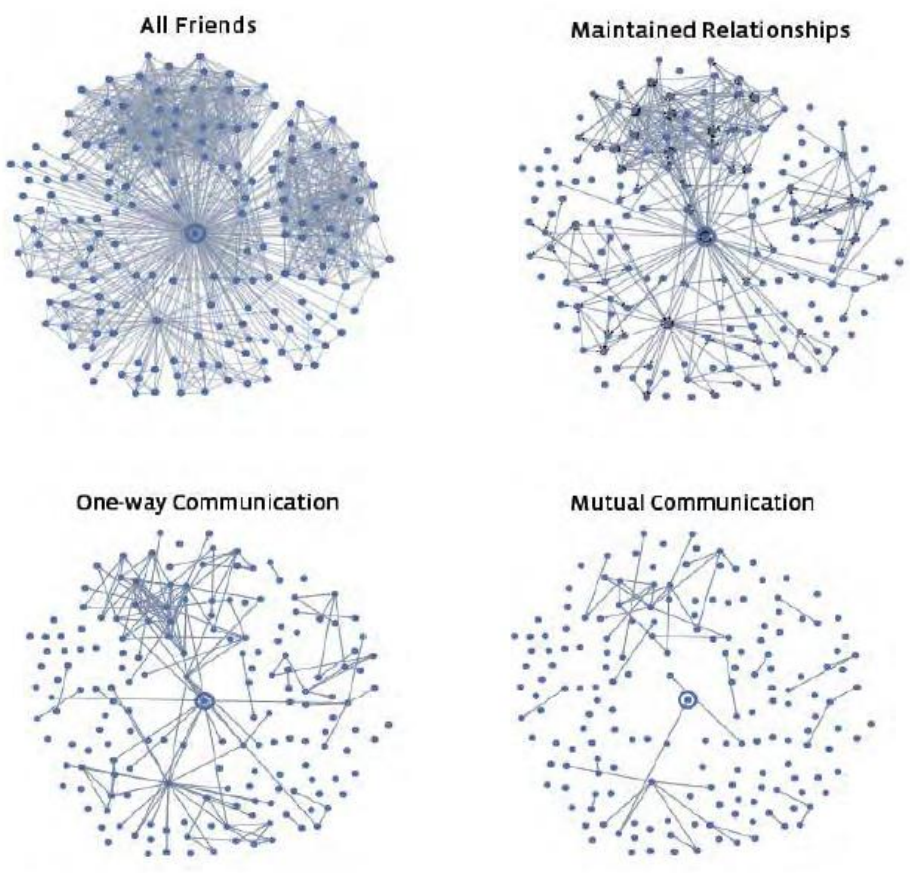


Tie Strengths

- How to measure strength:
  - Behavioral: intensity of interactions
  - Structural: Direct link, indirect link, embedded in a group, etc.
- Facebook Friendship



Source: EK (2010)

Are strong ties ‘local’?

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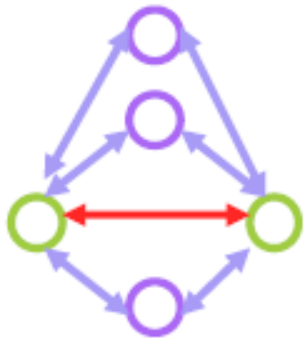
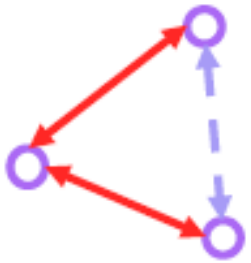
A strong tie

- frequent contact
- affinity
- many mutual contacts





“forbidden triad”:
strong ties are likely to “close”

Granovetter's (1973) Strength of Weak Ties

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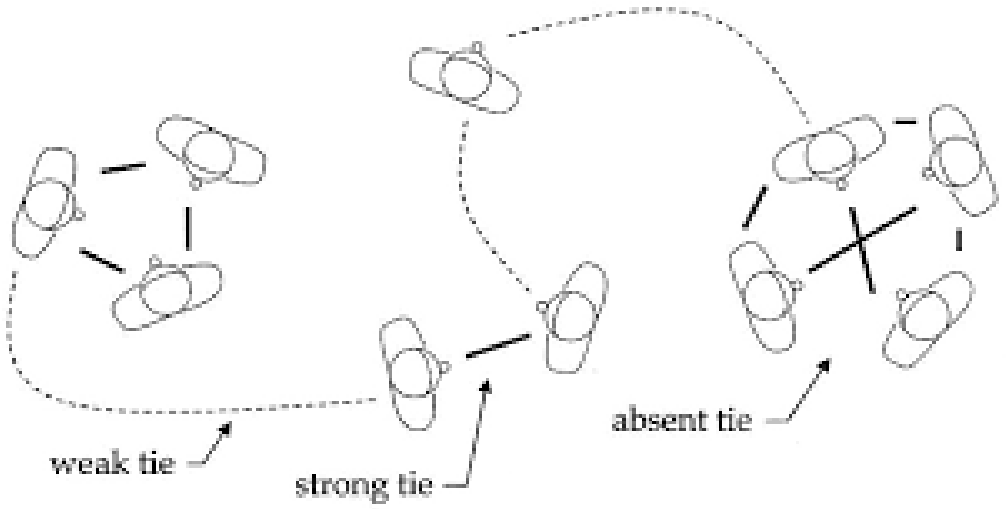
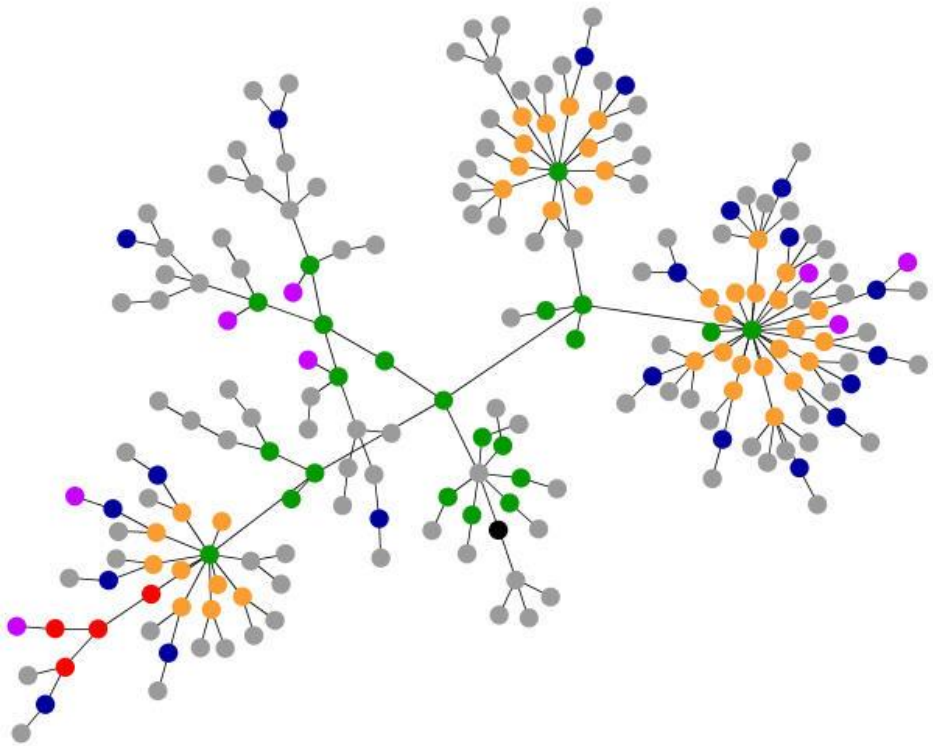
- Strong ties:
  - Close friends
  - Densely-knit network with many nodes mutually connected
- Weak ties:
  - Occasional acquaintances
  - Sparsely-knit network with many nodes independent from each other



Granovetter's (1973) Strength of Weak Ties



- The importance of weak ties
  - Weak tie is a crucial bridge between densely knit clumps of friends
  - Individual with too few weak ties will be deprived of latest information and heterogeneous ideas
  - Societies with too few weak ties will be fragmented and incoherent

Weak ties in Job Information

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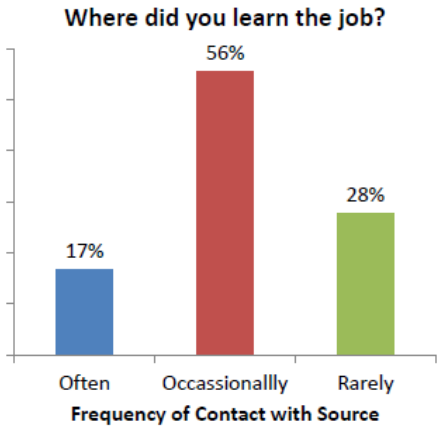
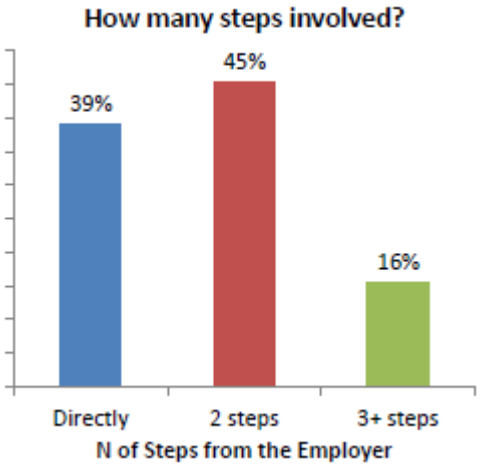


Source: Granovetter (1973).



- Granovetter's original 1973 research into the subject looked at how people find jobs. He discovered that information about jobs that led to employment was more likely to come from the weak ties with acquaintances than from closer friends.
- Occasional contact + short path

Bridges and Local Bridges

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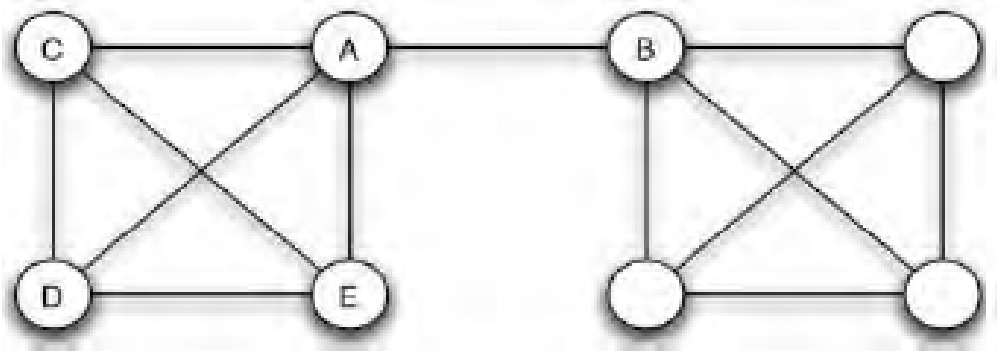


Figure 3.3: The A-B edge is a **bridge** because its removal would place A and B in two separate components.

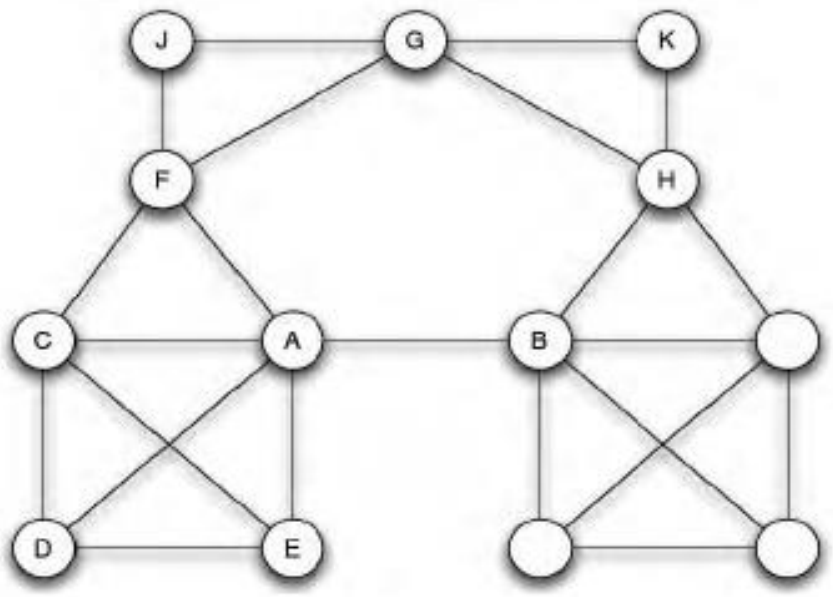
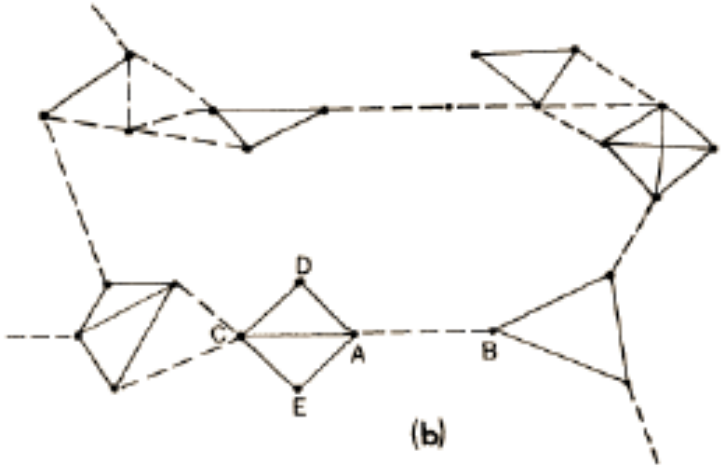


Figure 3.4: The A-B edge is a **local** bridge of span 4 because its removal would increase the distance between A and B to 4.

Bridges

- Strong ties are less likely to be a bridge or local bridge



Source: Granovetter, M. (1973). "The Strength of Weak Ties", American Journal of Sociology, Vol. 78, Issue 6, May 1973, pp. 1360-1380.

Strong Triadic Closure Property:
Strong ties are less likely to be a bridge

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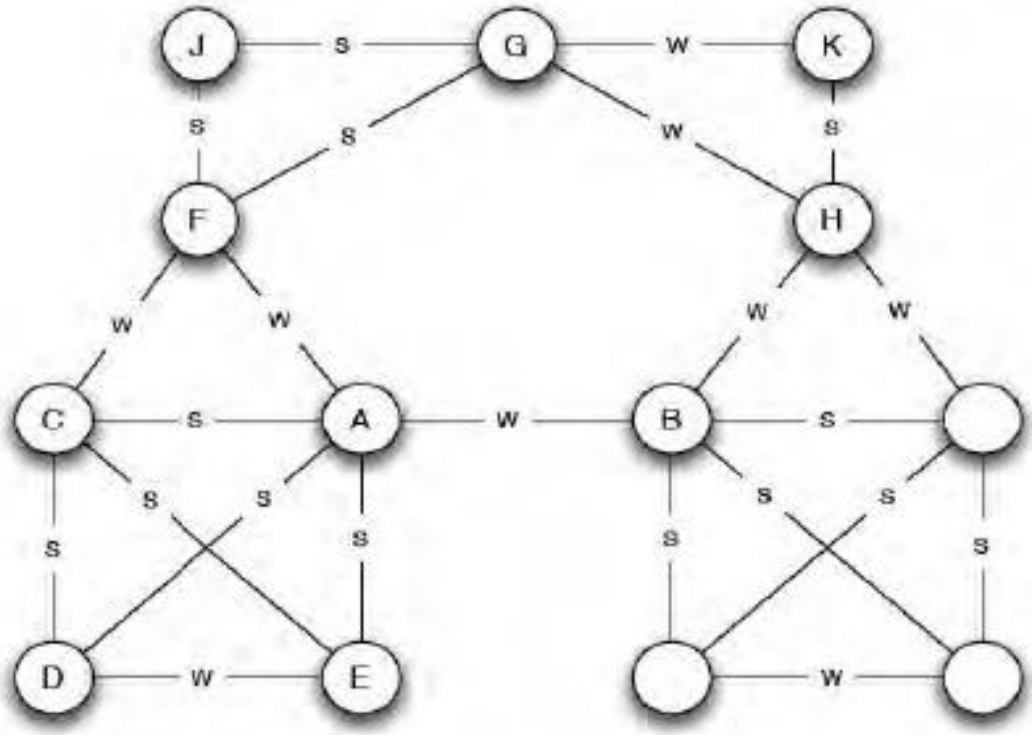
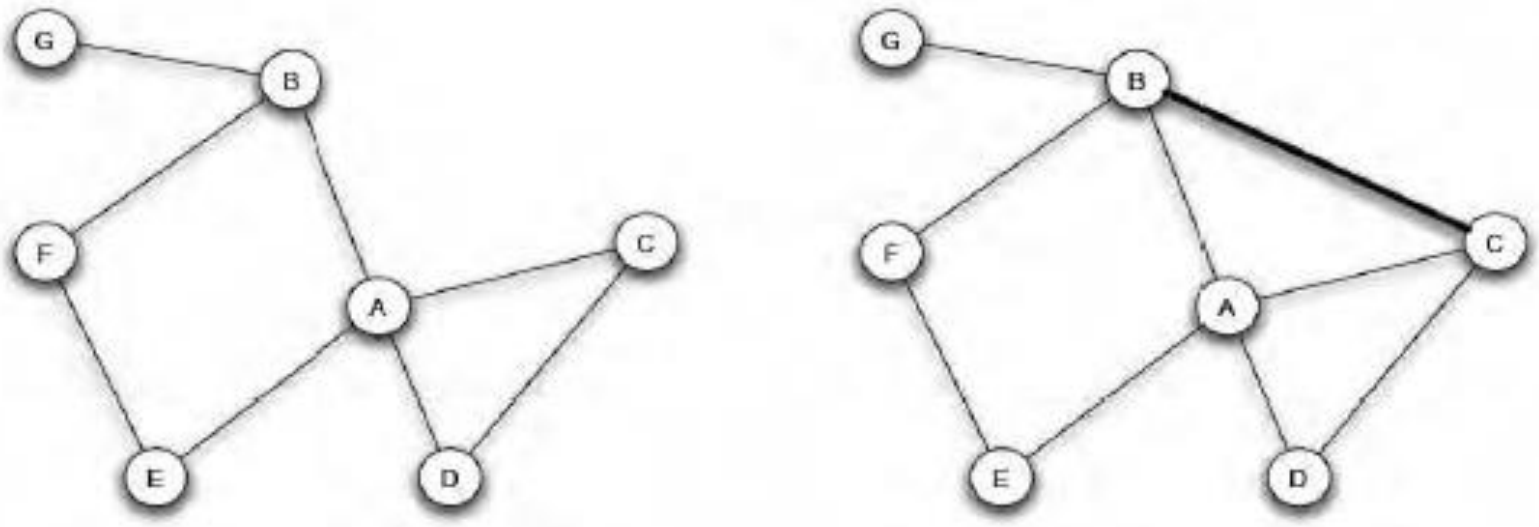


Node A would violate the property if it has strong ties with B and C but there is no tie (either strong or weak) at all between the latter

Figure 3.5: Each edge of the social network from Figure 3.4 is labeled here as either a *strong tie* (S) or a *weak tie* (W), to indicate the strength of the relationship. The labeling in the figure satisfies the Strong Triadic Closure Property at each node: if the node has strong ties to two neighbors, then these neighbors must have at least a weak tie between them.

Triadic Closure



(a) *Before B-C edge forms.* (b) *After B-C edge forms.*

Figure 3.1: The formation of the edge between B and C illustrates the effects of triadic closure, since they have a common neighbor A.

Why are friends (B and C) of a common friend (A) more likely to become friends themselves:
1. chances to meet each other; 2. similarity between them.

Triadic Closure as a Primary Drive for Tie Formation

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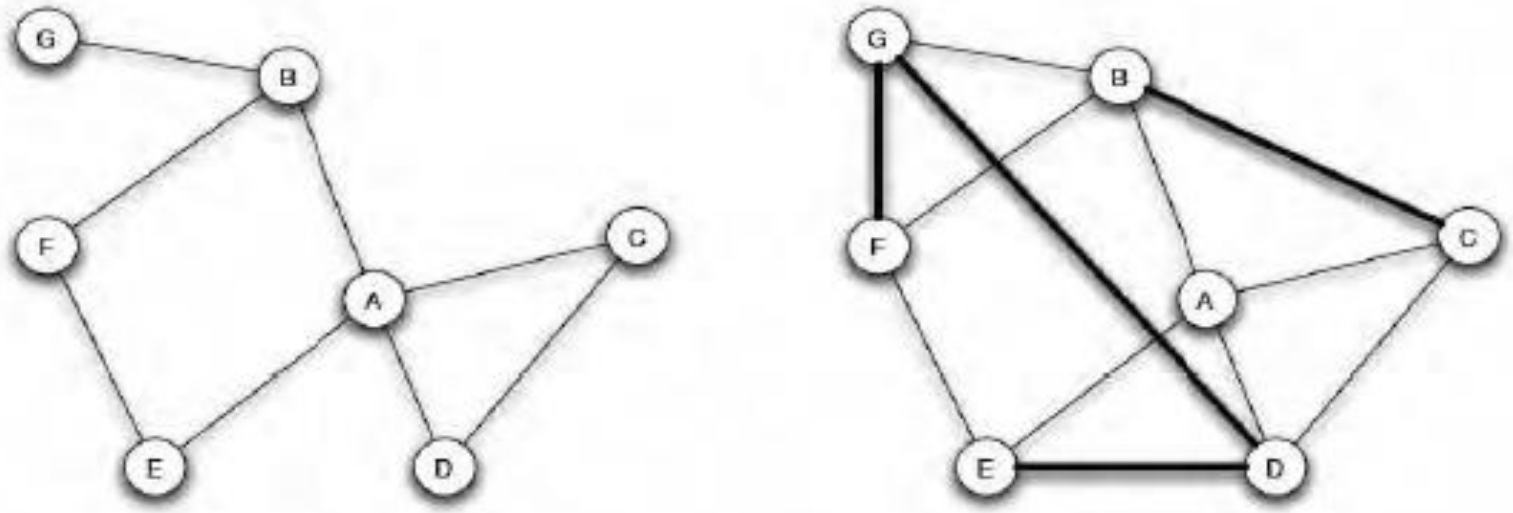
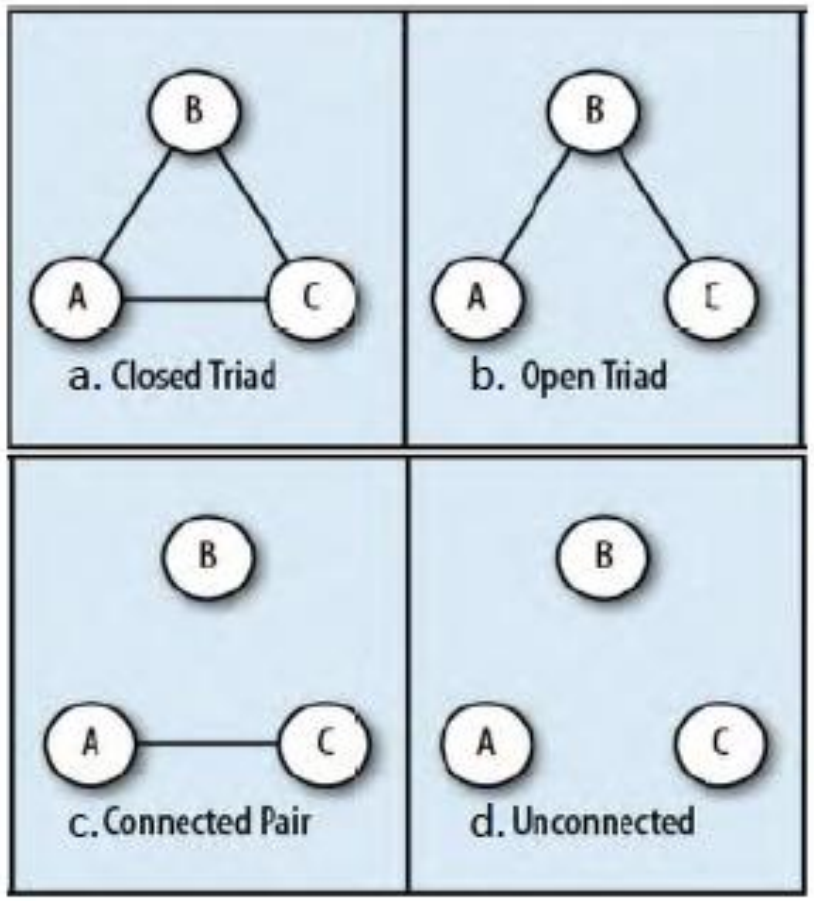


(a) *Before new edges form.* (b) *After new edges form.*

Figure 3.2: If we watch a network for a longer span of time, we can see multiple edges forming — some form through triadic closure while others (such as the *D-G* edge) form even though the two endpoints have no neighbors in common.

Edges of B-C and F-G are results of triadic closure but edge D-G is not.

Possible Triads of Undirected Graph



- Fig. a and Fig. b are two triads with a being closed and b open. Triads are used to calculate "clustering coefficient" (*CC*, more later).
- Fig. c and Fig. d are not triads and thus not counted in *CC*.
- Node B in Fig. b acts as a bridge or structural hole.

Clustering

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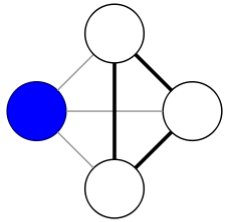
- Local clustering coeffcient (Watss & Strogatz, 1998)
- Duncan J. Watts and Steven Strogatz introduced the measure in 1998 to determine whether a graph is a small-world network.

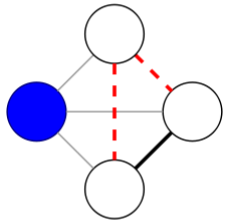
$$C_i = \frac{\text{\# of connections between i's neighbors}}{\text{max \# of possible connections between i's neighbors}}$$

$$C_{i\ \text{directed}} = \frac{\text{\# directed connections between i's neighbors}}{n_i * (n_i - 1)}$$

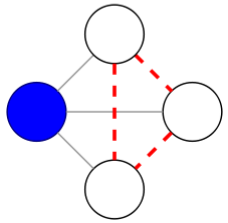
$$C_{i\ \text{undirected}} = \frac{\text{\# undirected connections between i's neighbors}}{n_i * (n_i - 1)/2}$$



c = 1



c = 1/3

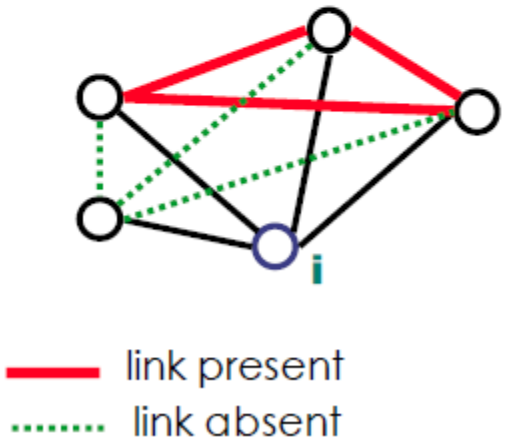


c = 0

Clustering

- Average over all n vertices

$$C = \frac{1}{n}\sum_i C_i$$



$n_i = 4$
max number of connections:
$4*3/2 = 6$
3 connections present
$C_i = 3/6 = 0.5$

Edge Embeddedness

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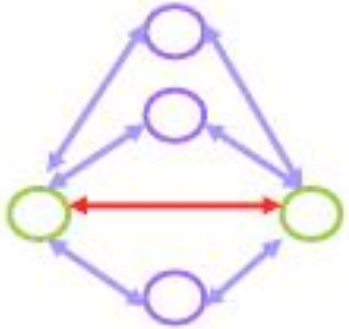
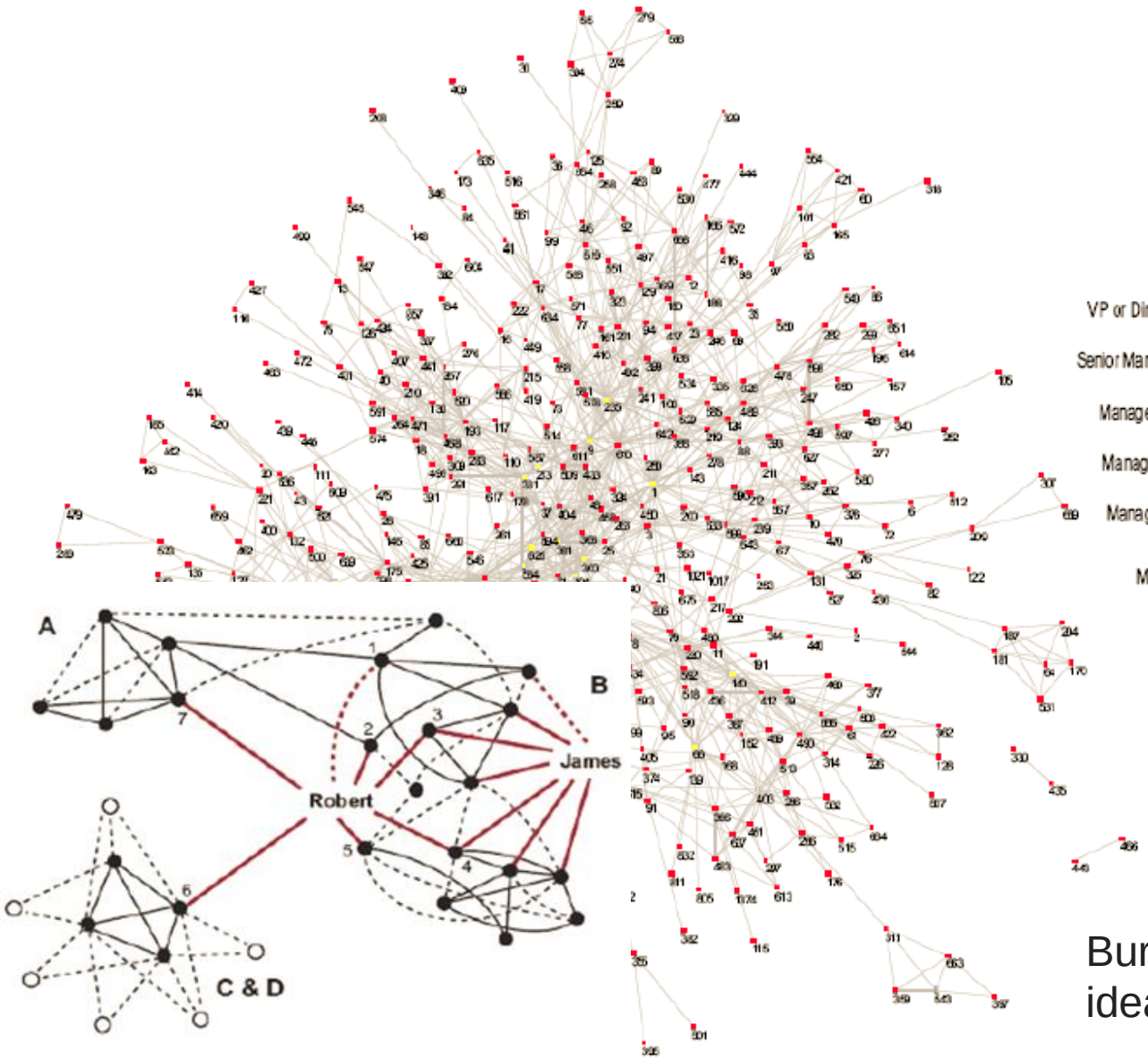
- embeddeness: number of common neighbors the two endpoints have



- neighborhood overlap:

$$\frac{\text{number of nodes who are neighbors of } \textit{both } A \text{ and } B}{\text{number of nodes who are neighbors of } \textit{at least one of } A \text{ or } B}$$



	Percent Social Isolates	Mean Network Size	Mean Network Constraint	Mean Number Cited as Discussion Partners	Mean Network Constraint Cited Discussn. Partners	Mean Path Distance (min-max) for the 476 connected managers in graph
VP or Director (25)	0%	12.6	29.8	4.9	70.2	3.3 (2.7-4.2)
Senior Manager (41)	5%	8.5	37.3	3.8	78.1	3.7 (2.9-6.4)
Manager III (121)	11%	6.4	50.2	3.7	77.9	4.0 (3.0-6.4)
Manager II (199)	27%	4.1	65.0	2.8	83.1	4.3 (2.8-6.4)
Manager I (287)	44%	3.4	73.6	2.4	83.4	4.6 (3.4-7.4)
Mean (673)	29%	5.0	60.5	2.9	81.0	4.2 (2.7-7.4)

Figure 2. Supply-Chain Discussion Network

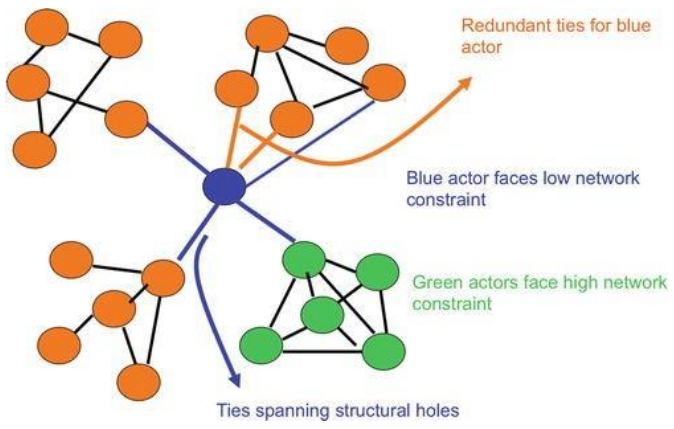
(excludes 193 social isolates)

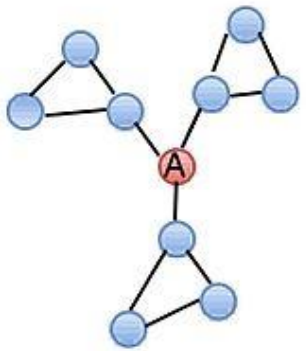
Burt, R. S. (2004). Structural holes and good ideas. *American journal of sociology*, *110*(2), 349-399.

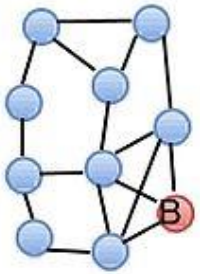
Source: Structural Holes and Good Ideas; R. Burt, American Journal of Sociology, 2004

Structural hole

- A structural hole refers to an “empty space” between contacts in a person's network. It means that these contacts do not interact closely (though they may be aware of one another). Actors on either side of the structural hole have access to different flows of information (see Fig. 1). Structural holes therefore reflect “an opportunity to broker the flow of information between people, and control the projects that bring together people from opposite sides of the hole” (Burt 2000).







Structural Holes and Good Ideas

- “I review evidence consistent with the hypothesis, then look at the networks around managers in a large American electronics company.”
- Network data were collected by the standard survey method of name generators and interpreters.
- The survey provides 5,010 observations of 4,139 relationships distinguishing five levels of connection between two people:
  - one person cited the other as someone with whom their idea was discussed and with whom supply-chain issues were frequently discussed
  - one cited the other as someone with whom he or she frequently discussed supply-chain issues, but not as someone with whom their idea was discussed
  - two people sometimes discussed supply-chain issues, but neither cited the other
  - one cited the other only as someone with whom their idea was discussed
  - neither person cited the other, and colleagues said the two people rarely discussed supply-chain issues

Findings

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- People who are networks bridge structural holes have
  - Higher compensation
  - Positive performance evaluations
  - More promotions
  - More good ideas
- These people are
  - More likely to express ideas
  - Less likely to have their ideas dismissed by judges
  - More likely to have their ideas evaluated as valuable

Final project

Networks are pluralistic: How can we conceptually categorize them?

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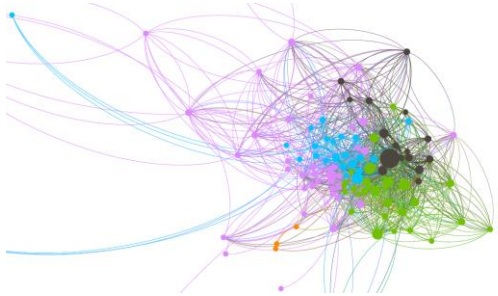
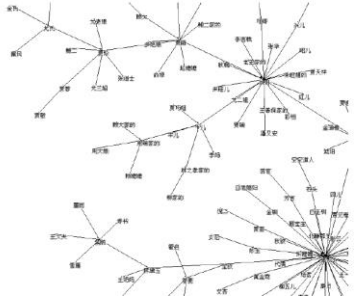
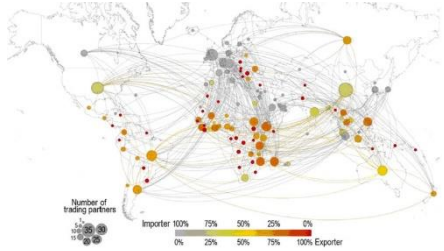
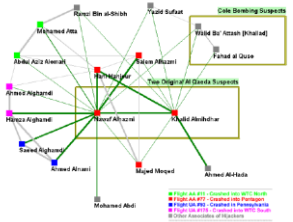
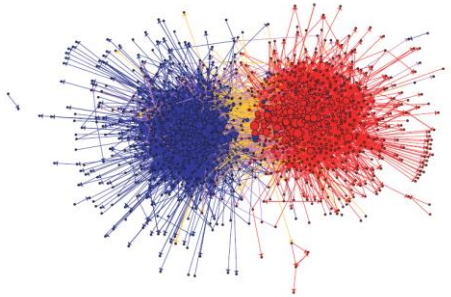


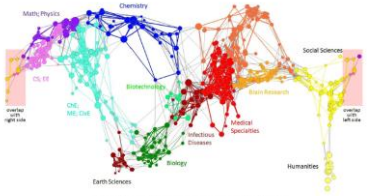




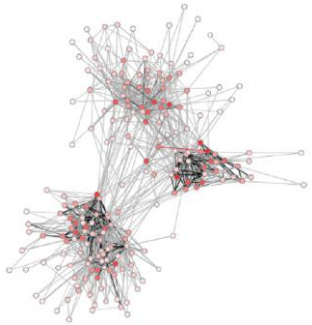


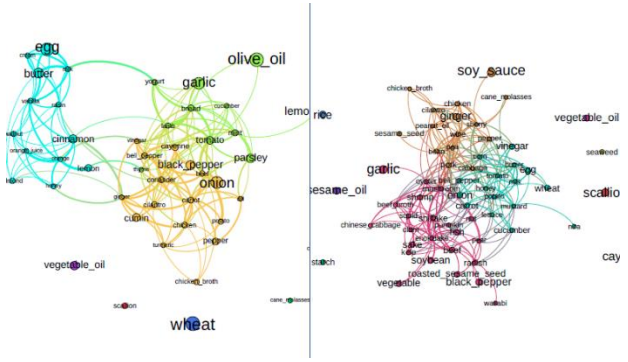
9/11 terrorist network
[Krebs, 2002]





Citation networks and Maps of science
[Börner et al., 2012]





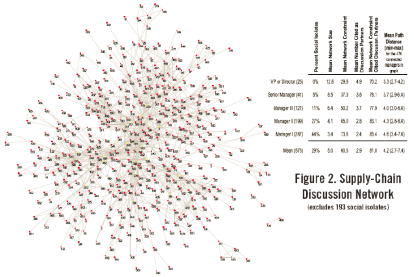


Figure 2. Supply-Chain Discussion Network
(excludes 193 social isolates)

Source: Structural Holes and Good Ideas; R. Burt, American Journal of Sociology, 2004

Categorization

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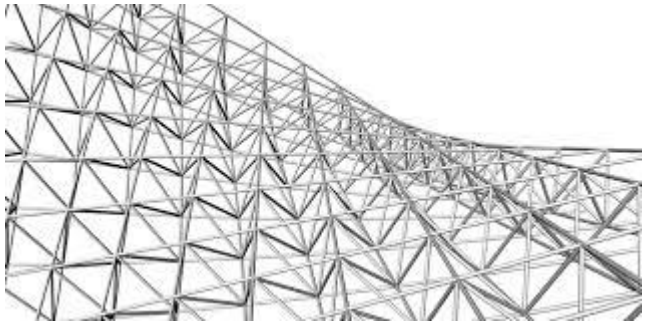

STRUCTURE NETWORK BEHAVIORAL NETWORK SEMANTIC NETWORK

Structure Networks
Key Concepts

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- Evolution of structural connections:
- Tie formation
- Tie maintenance
- Tie dissolution

- Mechanisms for structural connections:
- Homophily (similarity)
- Triadic closure (transitivity)
- Preferential attachment (popularity)
- Diversity (structural holes, curiosity)
- Etc.

Behavioral networks

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- Structural:
- A **prerequisite** for interactive activities in behavioral networks
- **Static**
- **Unweighted** (i.e., presence or absence of ties)
- Both **undirected** and **directed**

- Behavioral:
- Both **result** and **cause** of structural networks
- **Dynamic** (interactions on and off constantly)
- **Weighted** (i.e., frequency or intensity of interactions)
- Generally **directed**

Examples of Behavioral Networks

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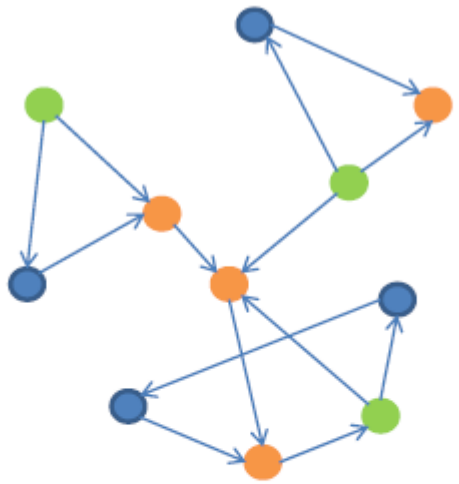
- Information Diffusion:
  - Sending/replying emails
  - Posting/commenting
  - Tweeting/retweeting
  - Resource sharing
- Other exchange activities:
  - Import/export
  - Selling/Buying
  - Borrowing/Lending
  - Attacking/Defensing
  - Virus/Disease transmission

Behavioral Network Key Concepts

- Information Cascades:
  - Information diffusion
  - Opinion leaders
  - Herding behavior
  - Sequential decision
- Questions to ask:
  - Who are more influential? Why?
  - Why are some messages diffused more quickly/widely than others?



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Opinion leaders in behavioral networks:

- Authorities (indegree ≥ 2)
- Hubs (outdegree ≥ 2)
- Followers (indegree & outdegree < 2)

Semantic Network

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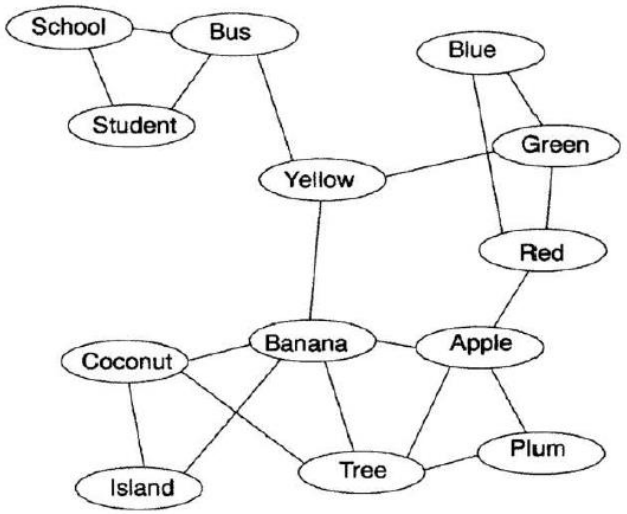
- What is semantics:
  - Semantics is about "meaning" of language
  - Language is made of words
  - Meaning of a word emerges from its association with other words

- Structural (link) network: who is connected to whom
- Behavioral (action) networks: who interacts with whom
- Semantic (meaning) networks: what shares similarity with what

Association of words



Meaning network

Co-occurrence Matrix Discover Related Phrases

	Starbucks Salted Carmel Mocha	Starbucks Coffee	Peppermint Mocha	Christmas Season	Fall Season
Starbucks Salted Carmel Mocha	1	0.8	0.3	0	0.5
Starbucks Coffee	0.8	1	0.8	0.2	0.2
Peppermint Mocha	0.3	0.8	1	0.5	0
Christmas Season	0	0.2	0.5	1	0
Fall Season	0.5	0.2	0	0	1

Phrases such as "Starbucks Coffee" and "Peppermint Mocha" are related by a high degree of co-occurrence.

These phrases become predicative of one another.

Co-occurrence network

Co-occurrence network

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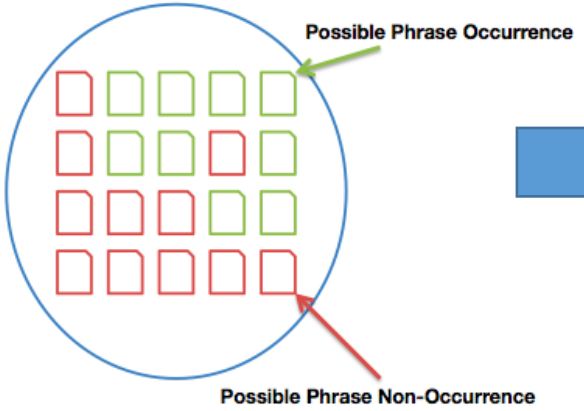
Co-occurrence of Phrases

The co-occurrence of phrases on pages in the document set can help discover related phrases.

Occurrence in Document Collection

Indexed set of documents is reviewed for occurrences of possible phrases.



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Food ingredients network is a semantic network

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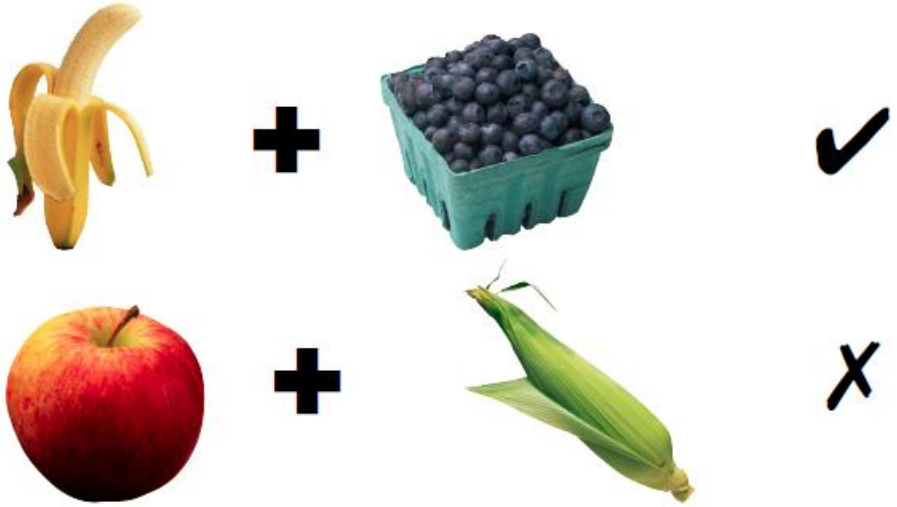
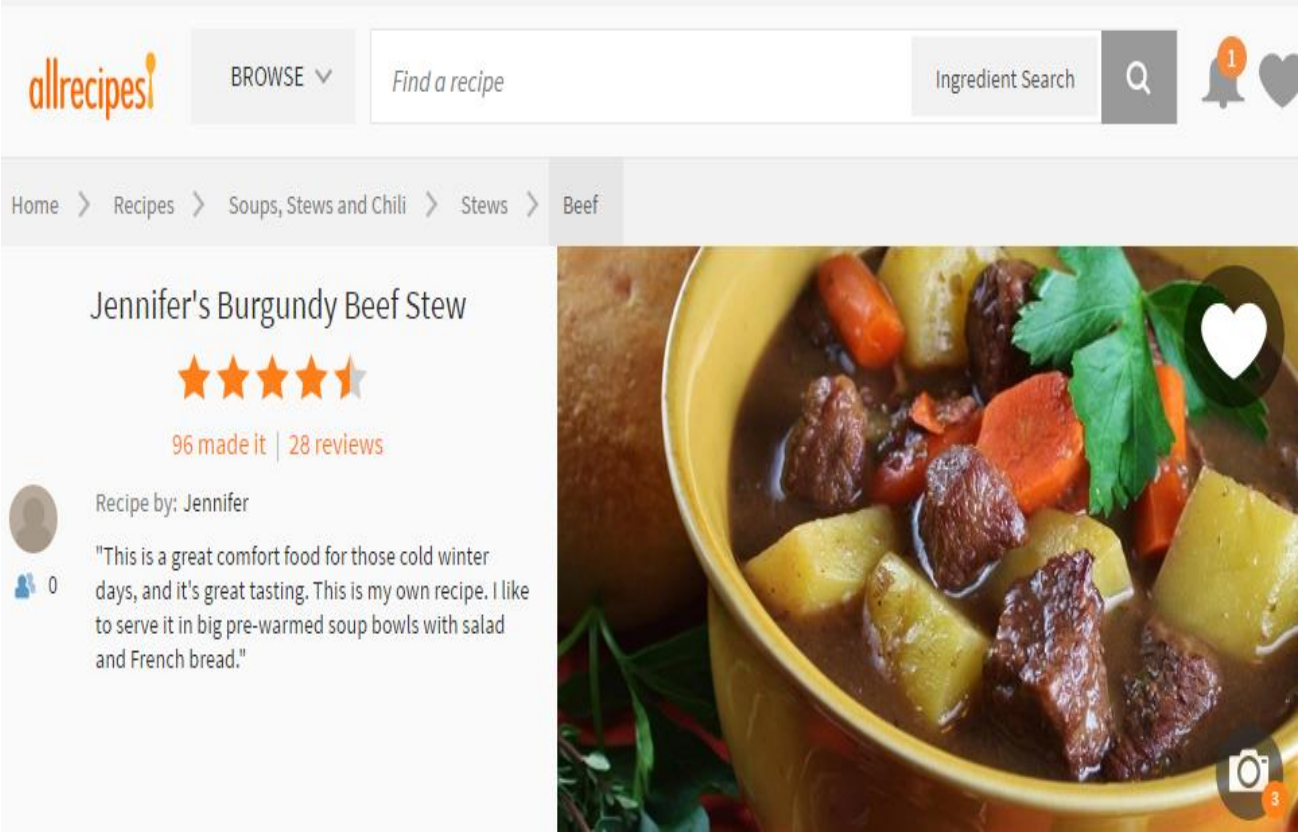


Recipe recommendation using ingredient networks

Chun-Yuen Teng[1], Yu-Ru Lin[2,3], Lada A. Adamic[1]

[1]School of Information, University of Michigan
[2]IQSS, Harvard University
[3]CCS, Northeastern University

Data collection from online



- Allrecipes.com
- 46,337 recipes
- Each recipe includes ingredients, nutrition info, cooking time, and regional information
- Review rating data for each recipe

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Ingredients

- 1 slice bacon
- 1 tablespoon olive oil
- 1/4 cup all-purpose flour
- 1 1/2 pounds London broil-cut beef, cut into chunks
- 2 cups low-sodium beef stock
- 1 cup Burgundy wine
- 4 carrots, cut into chunks
- 1 pound potatoes, cut into chunks
- 1/2 pound mushrooms, sliced
- 2 garlic cloves, minced
- 1 onion, cut into chunks
- 1 teaspoon dried marjoram
- 3/4 teaspoon ground thyme
- 1/2 teaspoon seasoned salt
- 1/8 teaspoon salt
- 1 pinch ground black pepper, or to taste

Add all ingredients to list

Construct a network

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- Nodes: ingredients
- Edges
  - undirected
  - Could be unweighted
  - Could be weighted as well – P(a,b)/P(a)*P(b)
  - Compromise – if weight >= 0.5 then connected, if weight < 0.5, not connected

Visualization

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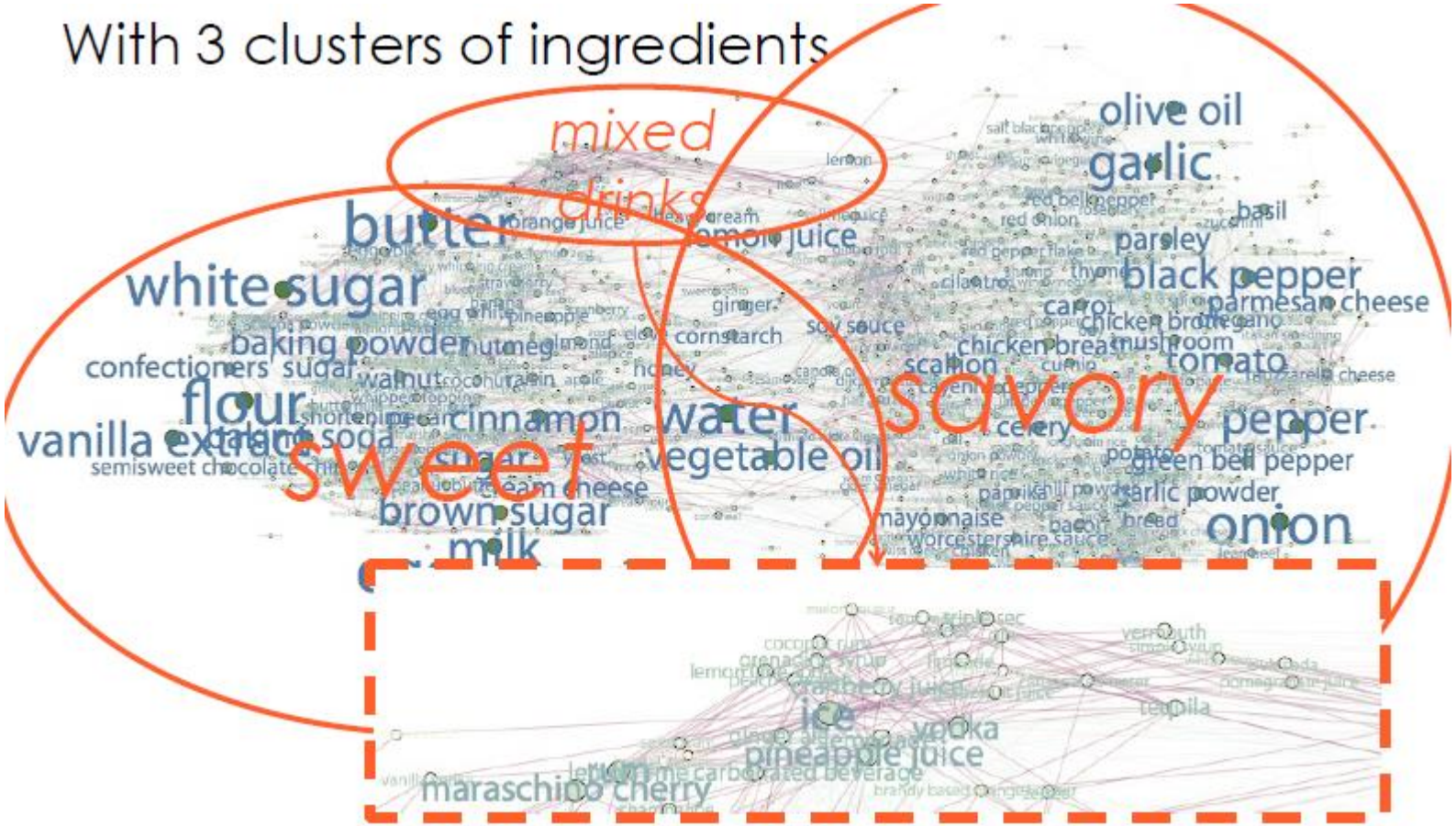

Guess the cuisine: sub-network using region as a divider

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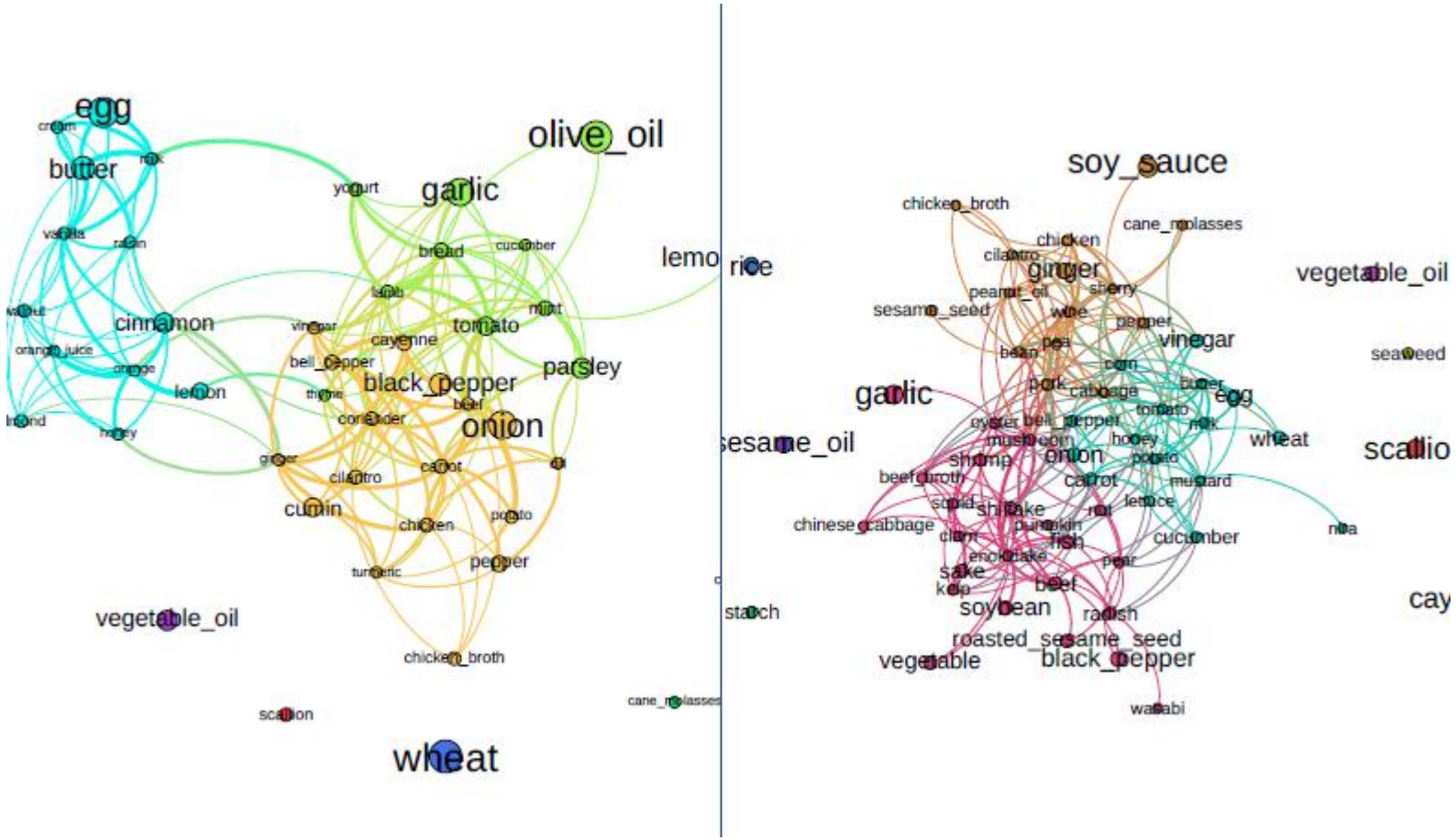

Lab 5

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- textual analysis example